PGR106
Linear Algebra

Lecture 6 – Norms & Basis Vectors

Kristiania University College

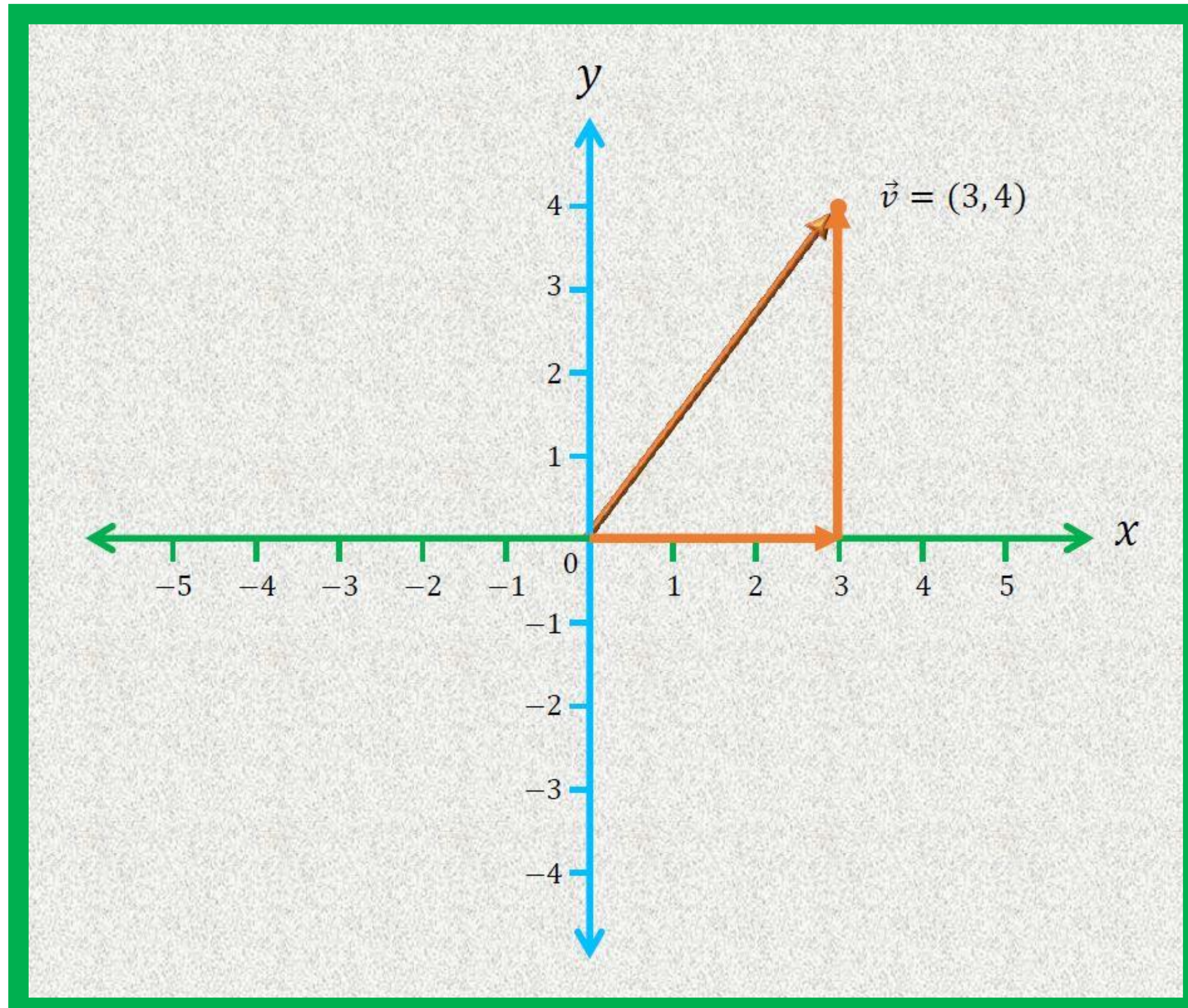
Lengths & Norms

Vectors

Direction

Magnitude (length)

$$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$



Lengths & Norms

- In Machine Learning (ML), the measure of the length of the vector is performed using a function known as the Norm.

- Different types of Norm:
  - L^2 Norm (Euclidean distance): it is the most widely used norm in ML.
  - L^1 Norm (Manhattan distance): it is used in ML when the difference between zero elements and non-zero elements of the vector is important.
  - L^p Norm (P-Norm): it is the generalization of L^1 and L^2 norms.
  - $L^{infinity}$ Norm (Infinity Norm): it is used in optimization problems where the goal is to minimize the maximum error or cost.

Lengths & Norms

- L^2 Norm

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$|v|_2 = |v| = \sqrt{{v_1}^2 + {v_2}^2}$$

$$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$|v|_2 = \sqrt{3^2 + 4^2}$$

$$|v|_2 = \sqrt{9 + 16}$$

$$|v|_2 = \sqrt{25} = 5$$

Lengths & Norms

- L^2 Norm

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \qquad |v|_2 = \sqrt{{v_1}^2 + {v_2}^2 + {v_3}^2}$$

$$\vec{v} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}$$

$$|v|_2 = \sqrt{3^2 + (-2)^2 + 5^2}$$

$$|v|_2 = \sqrt{9 + 4 + 25}$$

$$|v|_2 = \sqrt{38} = 6.164$$

Lengths & Norms

- L^2 Norm

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ . \\ . \\ . \\ v_n \end{bmatrix} \qquad |v|_2 = \sqrt{\sum_{i=1}^{n} {v_i}^2}$$

Lengths & Norms

- L^1 Norm

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$|v|_1 = |v_1| + |v_2|$$

$$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$|v|_1 = |3| + |4| = 7$$

Lengths & Norms

- L[1] Norm

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \qquad |v|_1 = |v_1| + |v_2| + |v_3|$$

$$\vec{v} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} \qquad |v|_1 = |3| + |-2| + |5| = 10$$

Lengths & Norms

- L^1 Norm

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ . \\ . \\ . \\ v_n \end{bmatrix} \qquad |v|_1 = \sum_{i=1}^{n} |v_i|$$

Lengths & Norms

- L^p Norm

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \cdot \\ \cdot \\ \cdot \\ v_n \end{bmatrix} \qquad |v|_p = \sqrt[p]{\sum_{i=1}^{n} |v_i|^p}$$

Lengths & Norms

- L^p Norm

$p = 4$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$|v|_4 = \sqrt[4]{|v_1|^4 + |v_2|^4}$$

$p = 4$ $\quad |v|_4 = \sqrt[4]{|v_1|^4 + |v_1|^4}$

$$\boxed{\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}}$$

$$|v|_4 = \sqrt[4]{|3|^4 + |4|^4}$$

$$|v|_4 = \sqrt[4]{81 + 256}$$

$$|v|_4 = \sqrt[4]{337} = 4.285$$

Lengths & Norms

- L$^{\text{infinity}}$ Norm

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ . \\ . \\ . \\ v_n \end{bmatrix} \qquad |v|_\infty = \max_i |v_i|$$

Lengths & Norms

- L^{infinity} Norm

$$|v|_\infty = \max_i |v_i|$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$|v|_\infty = \max(\,|v_1|, |v_2|\,)$$

$$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$|v|_\infty = \max(\,|3|, |4|\,)$$

$$|v|_\infty = 4$$

Lengths & Norms

- $L^{infinity}$ Norm

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \qquad |v|_\infty = \max(\,|v_1|, |v_2|, |v_3|)$$

$$\vec{v} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} \qquad |v|_\infty = \max(\,|3|, |-2|, |5|)$$

$$|v|_\infty = 5$$

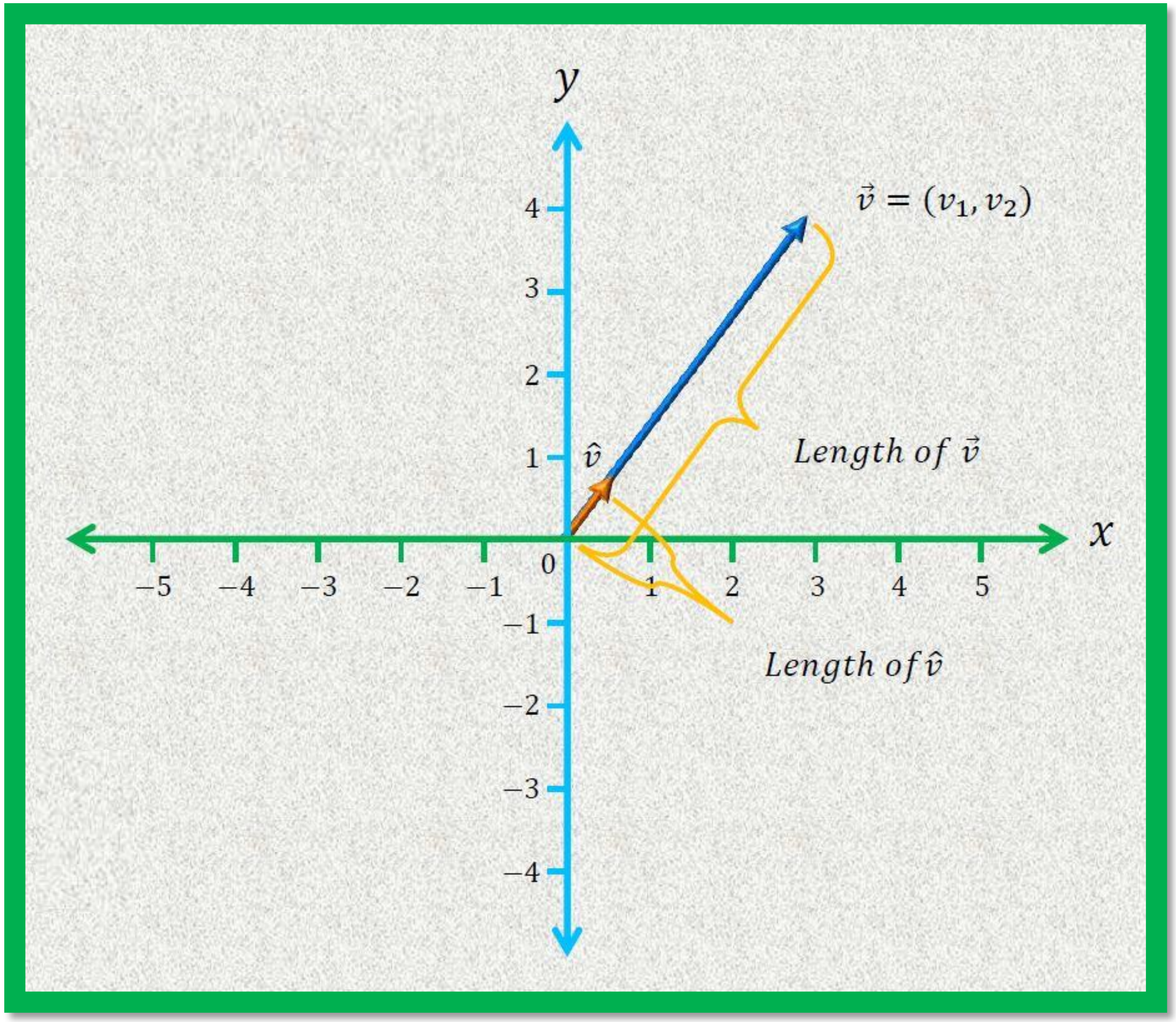
Unit Vectors

- A vector whose length is 1 is called a unit vector.

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$|v| = \sqrt{{v_1}^2 + {v_2}^2}$$

$$\hat{v} = \frac{v}{|v|} \qquad \hat{v} = \begin{bmatrix} v_1 \,/\, |v| \\ v_2 \,/\, |v| \end{bmatrix}$$



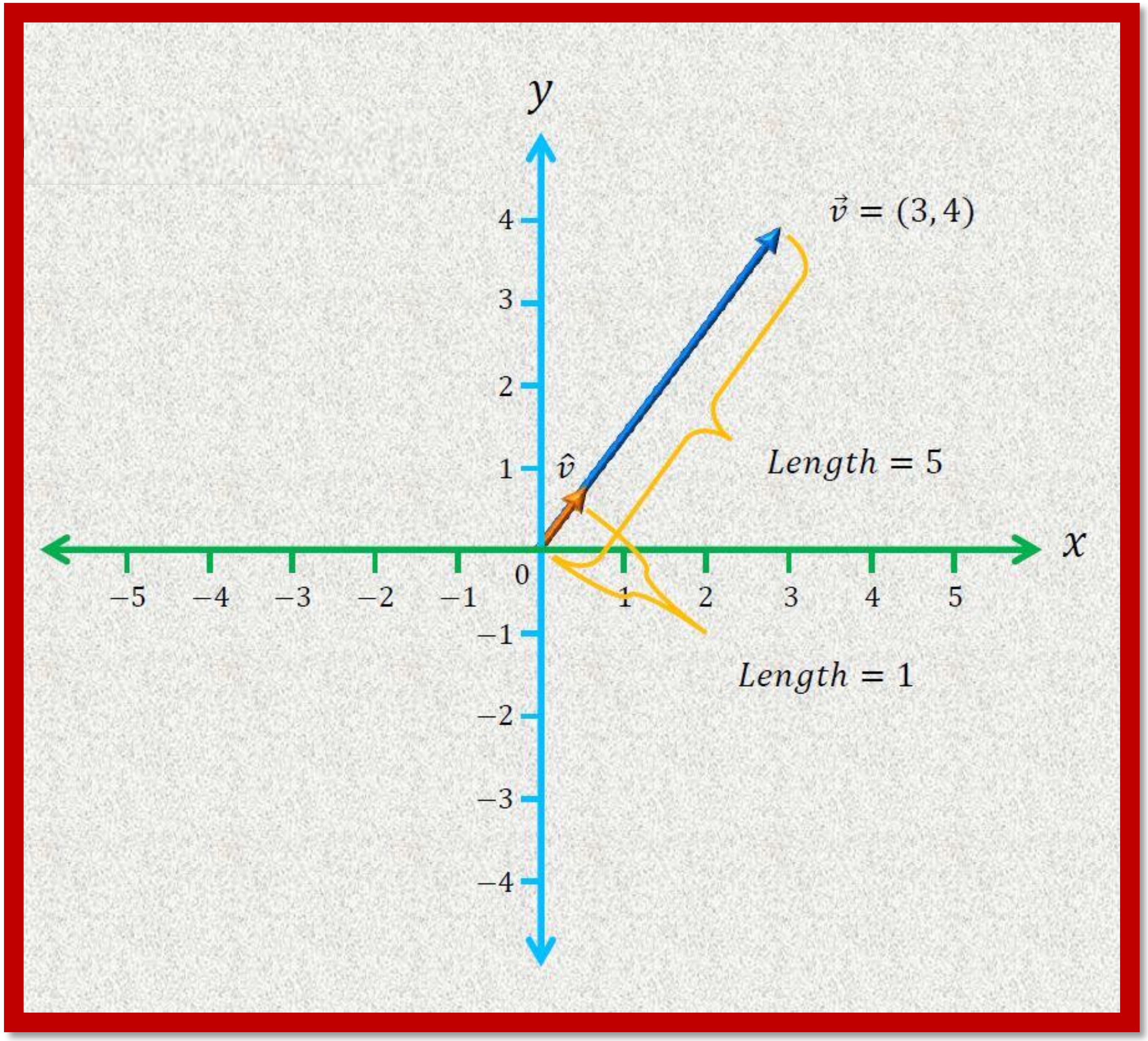
Unit Vectors

- A vector whose length is 1 is called a unit vector.

$$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$|v| = \sqrt{3^2 + 4^2} = 5$$

$$\hat{v} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$



Unit Vectors

$$\vec{v} = \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix}$$

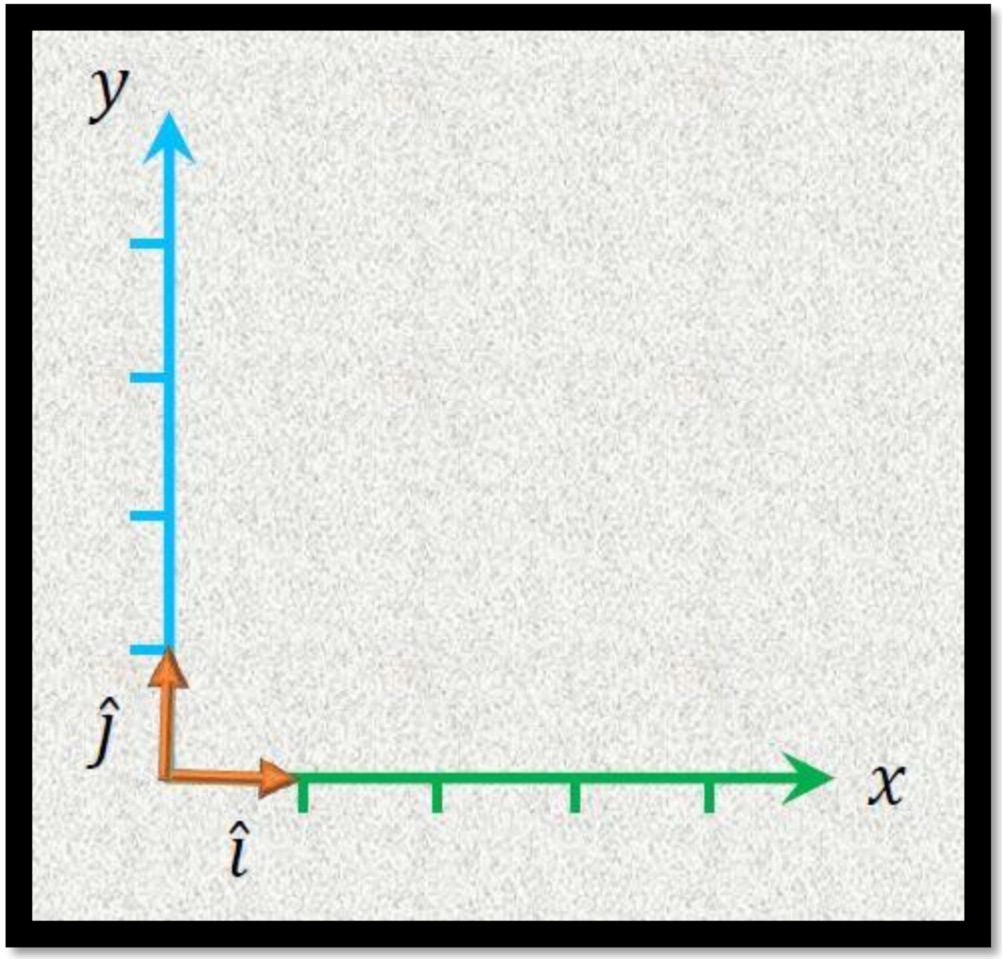
$$|v| = \sqrt{3^2 + (-7)^2 + 5^2} = \sqrt{83}$$

$$\hat{v} = \begin{bmatrix} 3/\sqrt{83} \\ -7/\sqrt{83} \\ 5/\sqrt{83} \end{bmatrix}$$

Standard Basis Vectors

- Standard basis vectors in $\mathbf{R}^2$ are special unit vectors denoted by $\hat{\boldsymbol{\imath}}$ $\hat{\boldsymbol{\jmath}}$

$$\hat{\imath} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \hat{\jmath} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

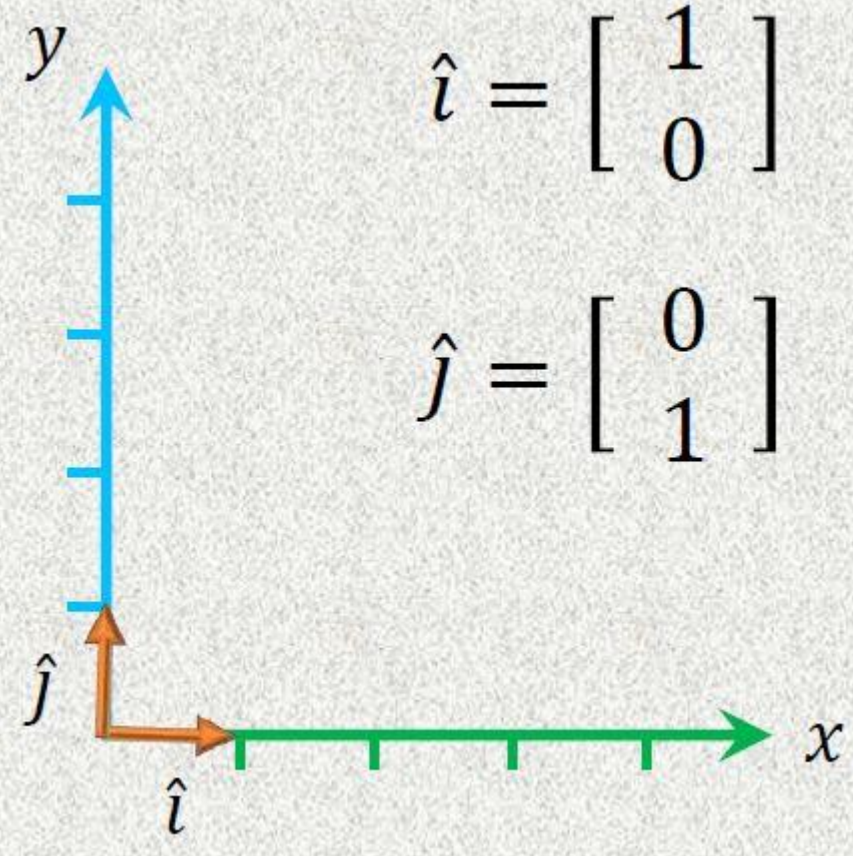


Standard Basis Vectors

$$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \qquad \underline{\vec{v} = 3\hat{\imath} + 4\hat{\jmath}}$$

$$\vec{v} = 3\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$



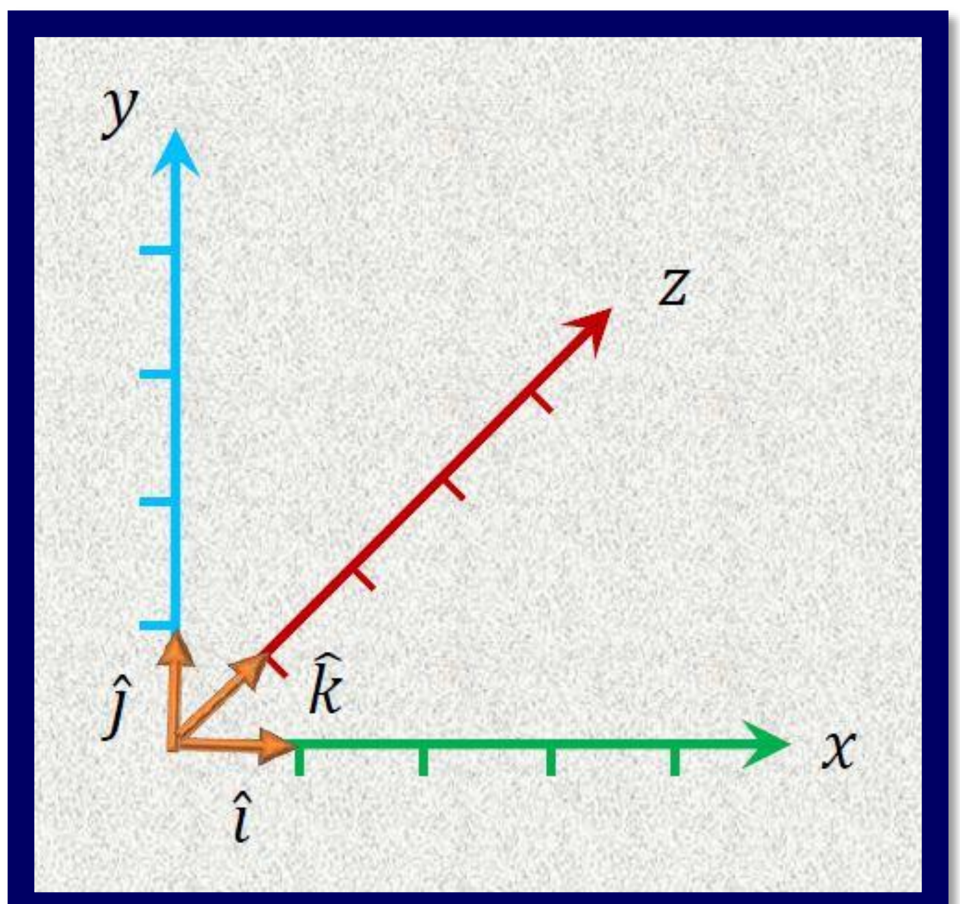
$$\hat{\imath} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\hat{\jmath} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Standard Basis Vectors

- Standard basis vectors in $\mathbf{R}^3$ are special unit vectors denoted by $\hat{\boldsymbol{\imath}}$ $\hat{\boldsymbol{\jmath}}$ $\widehat{\boldsymbol{k}}$

$$\hat{\imath} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad \hat{\jmath} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad \hat{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



Standard Basis Vectors

$$\vec{v} = \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix}$$

$$\vec{v} = 3\hat{\imath} - 7\hat{\jmath} + 5\hat{k}$$

$$\vec{v} = 3\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (-7)\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 5\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -7 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix}$$

$$\hat{\imath} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

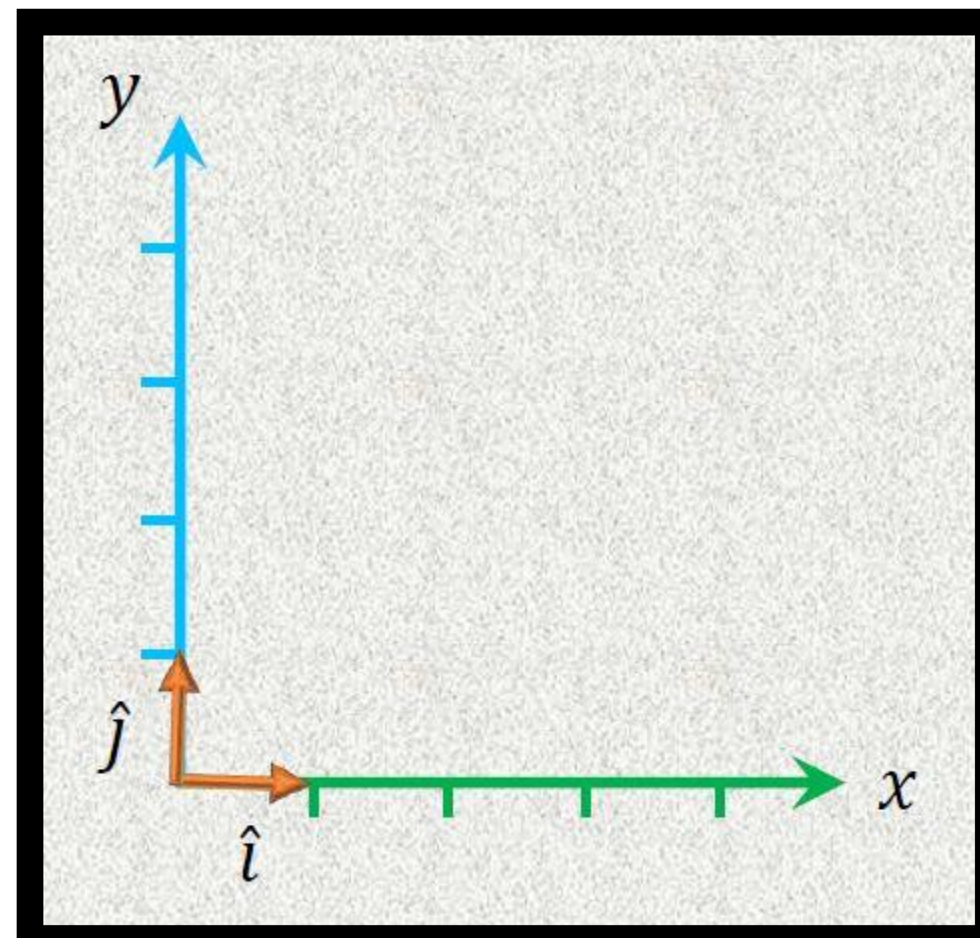
$$\hat{\jmath} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\hat{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Basis Vectors & Span

- Linear combination is the combination of addition and scalar multiplication and is significant because we want a particular combination (or all the combination).

$$\hat{\imath} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \hat{\jmath} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Basis Vectors & Span

- Can we have other basis vectors in R^2?

$$\hat{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \hat{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\vec{w} = c\hat{u} + d\hat{v}$$

$$\vec{w} = c\begin{bmatrix} 1 \\ 1 \end{bmatrix} + d\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} c \\ c \end{bmatrix} + \begin{bmatrix} d \\ -d \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} c+d \\ c-d \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$c + d = 3 \longrightarrow 1$$

$$c - d = 4 \longrightarrow 2$$

Basis Vectors & Span

$c + d = 3 \longrightarrow 1$

$4 + d + d = 3$

$2d = 3 - 4$

$2d = -1$

$d = -\frac{1}{2}$

$c - d = 4 \longrightarrow 2$

$c = 4 + d$

$c = 4 - \frac{1}{2}$

$c = \frac{7}{2}$

$c = \frac{7}{2}$ $\quad d = -\frac{1}{2}$

$$\vec{w} = \frac{7}{2}\hat{u} - \frac{1}{2}\hat{v}$$

Basis Vectors & Span

- Can we have other basis vectors in R^2?

$$\hat{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \hat{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\vec{w} = c\hat{u} + d\hat{v}$$

$$c = \frac{7}{2} \qquad d = -\frac{1}{2}$$

$$\vec{w} = \frac{7}{2}\hat{u} - \frac{1}{2}\hat{v}$$

Basis Vectors & Span

We can express $\boxed{\vec{w} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}}$

as a linear combination of $\boxed{\hat{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \hat{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}}$

$$\boxed{\vec{w} = \frac{7}{2}\hat{u} - \frac{1}{2}\hat{v}}$$

therefore, they are also basis vectors.

Basis Vectors & Span

- Basis for a vector space is the linearly independent vectors that span the space.

- n linearly independent vectors are needed to form the basis for vector space $\mathbf{R}^n$.

- For $\mathbf{R}^2$, we need **2** basis vectors which are linearly independent to span the entire $\mathbf{R}^2$ vector space.

Linear Independence

- A set of vectors $v_1, v_2, v_3 \ldots v_n$ is said to be Linearly Independent if no vector in the set can be expressed as a linear combination of other vectors.

Set of vectors $\{v_1, v_2, v_3 \ldots v_n\}$ is said to be Linearly Independent

if the equation $\boxed{c_1 v_1 + c_2 v_2 \ldots \ldots \ldots c_n v_n = 0}$

then the only solution to the equation is $c_1, c_2, \ldots c_n$ are all zero.

Linear Independence

- A set of vectors that are not linearly independent is said to be Linearly Dependent.

Set of vectors $\{v_1, v_2, v_3 \dots v_n\}$ is said to be Linearly Dependent

if the equation $$\boxed{c_1v_1 + c_2v_2 \dots \dots \dots c_nv_n = 0}$$

then $c_1, c_2, \dots c_n$ are not all zero (at least one coefficient is non-zero).

Linear Independence

$$u_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad u_2 = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$2u_1 = u_2$$

$$2\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

Linearly Dependent

$$v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad v_2 = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$$

$$cv_1 \neq v_2$$

$$2\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} \neq \begin{bmatrix} 4 \\ 9 \end{bmatrix}$$

$$3\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \end{bmatrix} \neq \begin{bmatrix} 4 \\ 9 \end{bmatrix}$$

Linearly Independent

Linear Independence

$$u_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad u_2 = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$c_1 u_1 + c_2 u_2 = 0$$

$$c_1 = 2 \qquad c_2 = -1$$

$$2\begin{bmatrix} 2 \\ 3 \end{bmatrix} - 1\begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Linearly Dependent

$$v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad v_2 = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$$

$$c_1 v_1 + c_2 v_2 = 0$$

$$c_1 = 0 \qquad c_2 = 0$$

$$0\begin{bmatrix} 2 \\ 3 \end{bmatrix} + 0\begin{bmatrix} 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Linearly Independent

Linear Independence – Basis Vector

- If a set of vectors v_1, v_2, v_3 … v_n contains more vectors than their components, then the set of vectors is said to be Linearly Dependent.

$$v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad v_2 = \begin{bmatrix} 4 \\ 9 \end{bmatrix} \quad v_3 = \begin{bmatrix} -1 \\ 3 \end{bmatrix} \quad v_4 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

Linearly Dependent

$$v_1 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \quad v_2 = \begin{bmatrix} 4 \\ 9 \\ 2 \end{bmatrix} \quad v_3 = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} \quad v_4 = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

Linearly Dependent

Linear Independence – Determinant

- If a set of vectors v_1, v_2, v_3 … v_n *is* represented as a matrix, it is said to be linearly dependent if the determinant of the matrix is zero.

- If the determinant of the matrix is non-zero, then it is said to be linearly independent.

- In other words, if the matrix is invertible, the columns are linearly independent.

- And if the matrix in non-invertible, then the columns are linearly dependent.

Linear Independence – Determinant

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 9 \end{bmatrix}$$

$$|A| = \ 2 * 9 - 4 * 3$$

$$= 18 - 12$$

$$= 6$$

Linearly Independent

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}$$

$$|A| = \ 2 * 6 - 4 * 3$$

$$= 12 - 12$$

$$= 0$$

Linearly Dependent

Linear Independence – Determinant

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 5 & 1 \\ 3 & 6 & 1 \end{bmatrix}$$

$$|A| = 1\begin{vmatrix} 5 & 1 \\ 6 & 1 \end{vmatrix} - 4\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} + 2\begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix}$$

$$|A| = \quad 1(5*1-1*6) - 4(2*1-1*3) + 2(2*6-5*3)$$

$$|A| = \quad 1(5-6) - 4(2-3) + 2(12-15) \quad = -1+4-6 \quad = -3$$

Linearly Independent

Linear Independence – Determinant

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 5 & 1 \\ 3 & 6 & 0 \end{bmatrix}$$

$$|A| = 1\begin{vmatrix} 5 & 1 \\ 6 & 0 \end{vmatrix} - 4\begin{vmatrix} 2 & 1 \\ 3 & 0 \end{vmatrix} + 2\begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix}$$

$$|A| = \quad 1(5 * 0 - 1 * 6) - 4(2 * 0 - 1 * 3) + 2(2 * 6 - 5 * 3)$$

$$|A| = \quad 1(-6) - 4(-3) + 2(12 - 15) \quad = -6 + 12 - 6 \ = 0$$

Linearly Dependent

Basis Vectors & Span

- Another Example

$$\hat{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \hat{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\vec{x} = \frac{3}{2}\hat{u} + \frac{1}{2}\hat{v}$$

$$\vec{x} = \frac{3}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix} + \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 3/2 + 1/2 \\ 3/2 - 1/2 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 4/2 \\ 2/2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Basis Vectors & Span

We can express $\vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

as a linear combination of $\hat{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \hat{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\vec{x} = \frac{3}{2}\hat{u} + \frac{1}{2}\hat{v}$$

therefore, they are also basis vectors.

Basis Vectors & Span

- Standard basis vectors can be used to express any vector using the linear combination.

- Other basis vectors can be used to express any vector using the linear combination.

- All the linear combinations of the basis vectors are known as **span**.

- Entire R^2 space is the span of the standard basis vectors.

- Entire R^2 space is the span of the basis vectors.

Basis Vectors & Span

- Span: any two <u>basis vectors</u> in R^2 can be used to express any vectors in R^2 space.

- These two basis vectors must be linearly independent in R^2.

- Span: any three <u>basis vectors</u> in R^3 can be used to express any vectors in R^3 space.

- These three basis vectors must be linearly independent in R^3.

Basis Vectors & Span

Exercise

Determine if the vectors (1, 1) and (1, −1) can be considered basis vectors for the R^2 vector space including the vector (4, 4).

End of Lecture 6